Unified Theoretical Model for Axially Loaded Concrete-Filled Steel Tube Stub Columns with Different Cross-Sectional Shapes

Siqi Lin[1]; Yan-Gang Zhao, M.ASCE[2]; Zhao-Hui Lu[3]; and Xi-Feng Yan[4]

Abstract: The existing theoretical models for the analysis of concrete-filled steel tube (CFST) columns are mainly limited to circular specimens. This paper is aimed at developing a unified theoretical model which is applicable to CFST columns with different cross-sectional shapes, including circular, square, rectangular and round-ended specimens. CFST columns with different shapes (original columns) were replaced with equivalent circular columns by assuming that they have the same sectional areas of core concrete and steel tube. The difference in the behavior between the original and equivalent CFST columns is attributed to the shape effect, which is indicated and modelled by a confinement effectiveness factor. Then, a unified theoretical model for axially loaded CFST columns with different cross-sectional shapes was developed based on an equivalent circular column with the shape effect considered. A total of 982 CFST columns, including 550 circular columns, 396 rectangular (and square) columns, and 36 round-ended columns, were collected to assess the proposed model. Results suggest that the proposed model is capable of accurately predicting the load-carrying capacity and load-deflection curves of CFST columns with different cross-sectional shapes. Finally, the proposed theoretical model was adopted to develop a unified design model of axial load-carrying capacity. **DOI: 10.1061/(ASCE)ST.1943-541X.0003150.** © *2021 American Society of Civil Engineers.*

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Introduction

The concrete-filled steel tube (CFST) column is a composite member. The external steel tube provides confinement for the infilled concrete, and the infilled concrete prevents or at least delays inward local buckling of the steel tube (Han 2002). The interaction between the steel tube and core concrete significantly improves the performance of CFST columns, thereby leading to their high strength and stiffness, favorable ductility and large energy absorption capacities. As a result of this superior behavior, CFST columns have gained increasingly popularity as applications in structural construction (Lin et al. 2018; Yan and Zhao 2020, 2021).

There are various cross-sectional shapes for CFST columns. Among them, circular columns, which generally provide good confinement to the core concrete and exhibit high strength, are widely used. Square or rectangular CFST columns, because of their easy connection with beams, large cross-section inertia, and aesthetic properties, are also popular applications in construction. Other shapes (e.g., round-ended rectangular, polygon and elliptical sections) are used as well for aesthetical reasons (Han et al. 2014).

During the past few decades, much experimental, numerical and theoretical research has been conducted to understand the behavior of CFST columns. Experimental investigations provide fundamental understanding of their behavior, but they are expensive and time-consuming (Tao et al. 2013). Thus, numerical methods [3D finite element (FE) or fiber beam element (FBE)] have commonly been adopted. Although numerical methods can precisely predict the behavior of composite structures, 3D FE models are relatively tedious and complicated to build, especially for the analysis of large structural systems, due to the complexity in the modelling and the need to rebuild the model and reinput the model information for each new case (Lin et al. 2020b). By contrast, FBE models are essentially empirical and cannot well simulate the interaction between the concrete and steel tube. The FBE method is developed based on fiber beam elements, which are made up of a series of segments along the element length with the cross section discretized into a large number of fibers (Monti and Spacone 2000). Nonlinear constitutive relations are assigned to each individual fiber and the sectional response is obtained through the integration of the stress-strain response of the cross-section fibers (Lin et al. 2020b). Compared with experimental and numerical investigations, the theoretical method, which can simulate the interaction between the concrete and the steel tube, and can potentially be used to predict the behavior of concrete confined by other materials, is considered versatile and powerful (Teng et al. 2007). Johansson (2002) developed a theoretical model for circular CFST columns which accounts for the passive confinement using an incremental procedure by continuously updating the confining stress. In Johansson's model, a volumetric strain history for the concrete is employed to model the dilation behavior of confined concrete. Similarly, Teng et al. (2013) proposed a theoretical model for circular carbon-fiber-reinforced polymer (CFRP)-confined CFSTs under axial compression, based on a newly-developed dilation model of confined

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concrete, and Lai and Ho (2016) developed a theoretical dilation model for unconfined and confined circular CFST columns, based on their experimental results. Lin et al. (2020a) developed an analytical model for axially loaded circular normal, high, and ultra-high strength concrete-filled steel tube stub columns, considering the influence of confining stress path. Choi and Xiao (2010) established an analytical model of axially compressed circular CFRP-confined CFST columns based on the equilibrium condition, deformation compatibility, and the stress-strain models of the constitutive materials.

It is worth noting that the foregoing theoretical models all focused on circular CFST columns. This paper is aimed at veloping a unified theoretical model that is applicable to CFST columns with different cross-sectional shapes. The CFST columns examined in this paper have circular, square, rectangular, and round-ended shapes. These columns are then replaced by equivalent circular columns, as described in the next section. Based on the established equivalent circular column, a unified theoretical model by considering the shape effect is proposed for CFST columns with different cross-sectional shapes. Then, the proposed theoretical model is adopted to develop a unified model for the design of axial load-carrying capacity. Finally, conclusions are drawn.

Equivalent Circular Column Considering Shape Effect

Equivalent Circular Column

CFST columns with circular, square, rectangular, and round-ended shapes can be considered as special cases of a rectangular CFST column with corner radii (typical form). The typical column and corresponding special cases are shown in Fig. 1, where b and h are the width and depth of the cross section, respectively; r_0 and r_1 are the corner radii of the outer side and inner side of the steel tube, respectively; and t is the thickness of the steel tube. In this paper, the width b is assumed not smaller than the depth h, i.e., $b \geq h$. From the figure, it can be seen that the typical column with $r_0 = r_1 = 0$ would turn into a rectangular column, and that with $r_0 = h/2$ and $r_1 = h/2 - t$ it would be a round-ended column. Additionally, the case of $r_0 = r_1 = 0$ and $h = b$ is the square column, and that of $r_0 = h/2$, $r_1 = h/2 - t$, and $h = b$ is the circular column. The typical CFST columns and the corresponding special ones in Fig. 1, named as original columns, are replaced by equivalent circular columns. The unified theoretical model is then developed based on the established equivalent circular column by introducing the shape effect to consider the different behaviors between the equivalent and original columns.

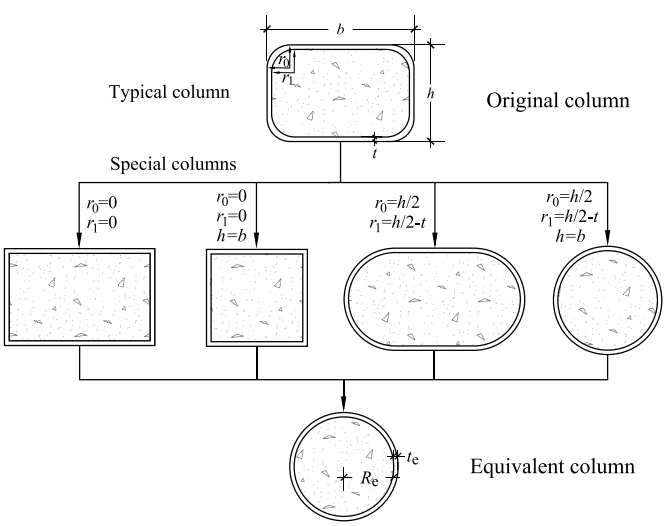

Fig. 1. Equivalent circular column.

The equivalent circular column concept had been adopted in several studies, e.g., Légeron and Paultre (2003) and Lam and Teng (2003). It should be noted that the equivalent circular specimens established in these studies generally have different cross-sectional areas from the original specimens. Hence, the different behavior between the equivalent and original specimens cannot be completely explained by the shape effect, Han et al. their difference may also result from the different sectional areas of the columns. Additionally, these studies focused on reinforced concrete or FRP-confined concrete columns. While the external reinforcement of such structures only provides lateral confinement, that of CFST columns, i.e., the steel tube, also bears the axial load.

In this paper a new establishment method of equivalent columns is proposed for CFST columns. The equivalent circular column is assumed to have the same sectional areas as the steel tube and core concrete with the original column, so that the different behaviors between the original and equivalent columns can be completely determined by the shape effect. A unified theoretical model is then developed based on the established equivalent circular column by considering the shape effect. Based on the above discussion, we have

$$A_c = (b - 2t)(h - 2t) - (4 - \pi)r_1^2 = \pi R_e^2 \tag{1}$$

$$A_s = 2t(b + h) - 4t^2 - (4 - \pi)(r_0^2 - r_1^2) = \pi[(R_e + t_e)^2 - R_e^2] \tag{2}$$

where R_e and t_e are the radius of the core concrete and the thickness of the steel tube in an equivalent circular column, respectively, and can be derived as follows:

$$R_e = \sqrt{\frac{(b - 2t)(h - 2t) - (4 - \pi)r_1^2}{\pi}} \tag{3}$$

$$t_e = \frac{\sqrt{bh - (4 - \pi)r_0^2} - \sqrt{(b - 2t)(h - 2t) - (4 - \pi)r_1^2}}{\sqrt{\pi}} \tag{4}$$

Based on Fig. 1, to assure that the typical CFST column can be transformed into rectangular and circular CFST columns, the relationship between the inner and outer radii should satisfy two boundary conditions, i.e., if $r_0 = 0$, $r_1 = 0$, and if $r_0 = h/2$, $r_1 = h/2 - t$. Then, their relationship can be given as follows, based on linear regression:

$$r_1 = \frac{h - 2t}{h} r_0 \tag{5}$$

Using Eqs. (3)–(5), the radius of the core concrete and the thickness of the steel tube in an equivalent circular CFST column can be obtained.

Shape Effect

The theoretical model for CFST columns with different cross-sectional shapes (original columns) was developed based on the established equivalent circular column. Because the equivalent CFST column is assumed to have same sectional areas of steel tube and

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Table 1. Details of CFST columns designed to study lateral confining stress

b (mm)	h (mm)	H (mm)	t (mm)	r_1 (mm)	f_y (MPa)	f_c (MPa)
670	670	2,010	4	85	350	48.3
670	420	2,010	4	85	350	48.3
670	230	2,010	4	85	350	48.3

core concrete as the original one, their differences can be attributed to the influence of different shapes, i.e., shape effect.

Different from circular CFST columns, the core concrete in CFST columns with other cross-sectional shapes experiences different lateral confinement effects, consequently leading to different behavior of the core concrete and columns. Hence, to well consider the shape effect on the behavior of confined concrete, the lateral confining stress should be clearly understood. A 3D FE model, built using the commercial software Abaqus version 6.14-5, was used to study the confining stress of core concrete in typical CFST columns. The concrete model of Tao et al. (2013) and the steel model of the authors (Lin and Zhao 2019) were adopted for the FE model. Although both the geometric conditions and the physical parameters of the column influence the lateral confining stress of the core concrete, it should be noted that geometric conditions influence the distribution and magnitude of lateral confining stress, whereas physical parameters mainly influence its magnitude. Because this section concerns the shape effect on the distribution of lateral confining stress, only the influence of the geometric conditions is discussed herein. Three typical CFST columns with different b/h ratios, which cover the most commonly encountered cases, were designed for the simulation analysis, details of which are summarized in Table 1. In the table, H is the height of the column, taken as three times the width; f_y is the yield strength of steel tube, and f_c is the compressive strength of unconfined cylindrical concrete. Because the specimens in Table 1 were designed to investigate the influence of cross-sectional shape rather than material properties on the distribution of confining stress of core concrete, the material properties, i.e., f_y and f_c, were taken as constants, which were set arbitrarily for the designed columns.

The distributions of lateral confining stress at the state of peak load of the specimens with different b/h ratios under the cylindrical coordinate system are shown in Fig. 2. It can be observed that the lateral confining stresses near the corners are significant, whereas those at the middle of the width and height are insignificant, suggesting that the concrete near the round corner is effectively confined, whereas that at the middle of the width and height is ineffectively confined. Similar results were found for square FRP-confined concrete columns by Lin and Teng (2020). Based on the results in the figure, the zone where the confining stress developed fully due to the confinement at the round corner and the arching action at midway is considered to be effectively confined, whereas the remaining zones are ineffectively confined. It is commonly assumed that the arching action occurs in the form of a second-degree parabola (Mander and Priestley 1988). This assumption is also adopted in this paper. Then, the effectively and ineffectively confined concrete zones can be seen in Fig. 3, where A_{ce} denotes the area of the effectively confined zone; and A_{hi} and A_{bi} denote the areas of ineffectively confined concrete along the height and width, respectively.

The shape effect influences the confinement effectiveness, which can be indicated by a confinement effectiveness factor k_e, defined as the ratio of the area of effectively confined concrete to the total sectional area of the core concrete

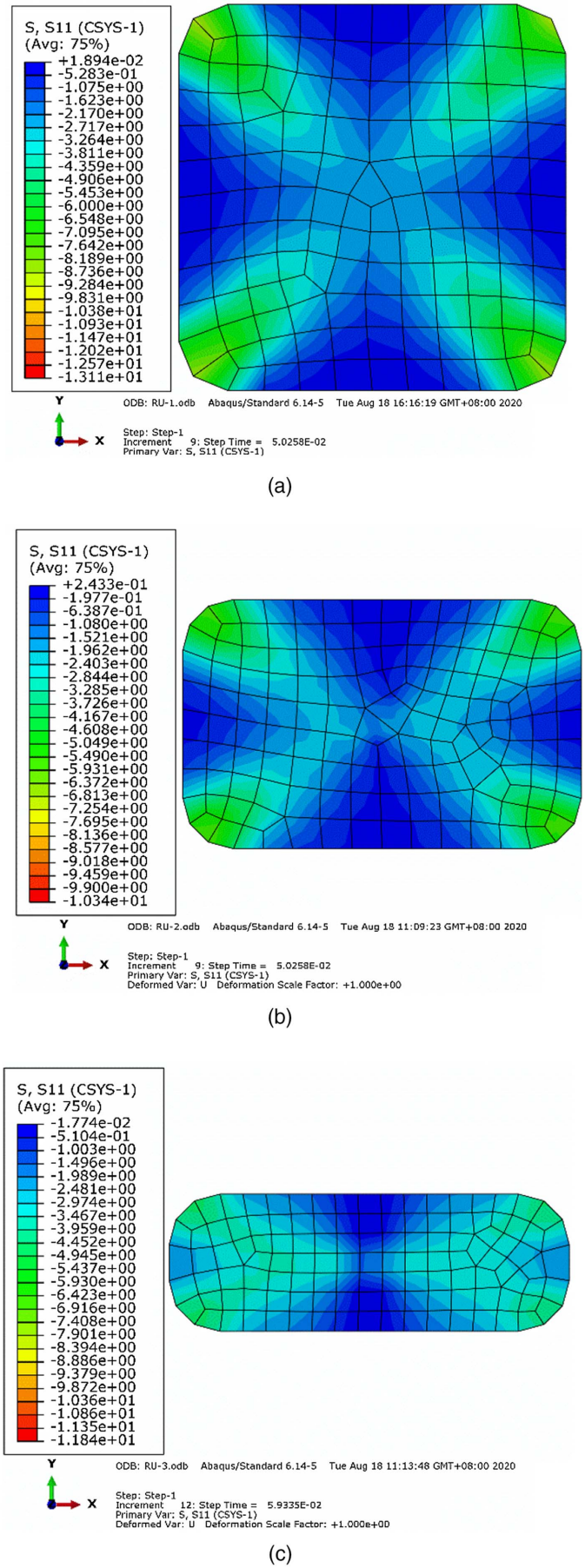

Fig. 2. Distribution of confining stress under cylindrical coordinate system: (a) $b/h = 1.0$; (b) $b/h = 1.6$; and (c) $b/h = 2.9$.

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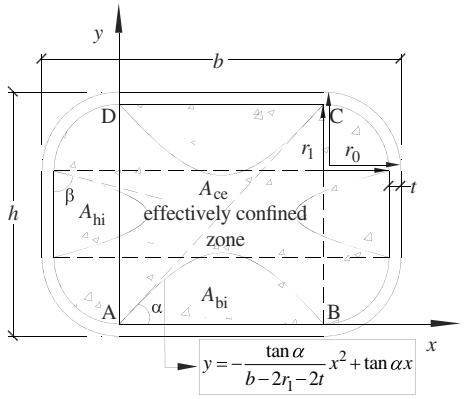

Fig. 3. Effectively and ineffectively confined concrete in a typical CFST column.

$$k_e = \frac{A_{\text{ce}}}{A_c} \tag{6}$$

Based on Fig. 3, A_{ce} can be determined by

$$A_{\text{ce}} = A_c - 2(A_{\text{hi}} + A_{\text{bi}}) \tag{7}$$

Eqs. (6) and (7) suggest that the effectiveness factor can be obtained once the areas of ineffectively confined concrete along the height and the width are determined.

Assuming the parabola along the width has an initial slope of α, the parabola can be obtained as

$$y = -\frac{\tan\alpha}{b-2r_1-2t}x^2 + x\tan\alpha \tag{8}$$

Then, we have

$$A_{\text{bi}} = \int_0^{b-2r_1-2t}\left(-\frac{\tan\alpha}{b-2r_1-2t}x^2 + x\tan\alpha\right)dx = \frac{\tan\alpha}{6}(b-2r_1-2t)^2 \tag{9}$$

Similarly, the area of ineffectively confined concrete along one side of the height can be determined as

$$A_{\text{hi}} = \frac{\tan\beta}{6}(h-2r_1-2t)^2 \tag{10}$$

where β = initial slope of the parabola along the height.

Substituting Eqs. (1), (7), (9), and (10) for Eq. (6) leads to

$$k_e = 1 - \frac{\tan\alpha(b-2r_1-2t)^2 + \tan\beta(h-2r_1-2t)^2}{3(b-2t)(h-2t) - 3(4-\pi)r_1^2} \tag{11}$$

In previous studies, the parabola was commonly assumed to have an initial slope of 45° (Mander and Priestley 1988; Lam and Teng 2003). However, based on the results in Fig. 2, simply assuming the initial slope to be 45° seems unreasonable for a column with a larger b/h. The analysis of the FE model (Fig. 2) suggests that the tangent line of the parabola at the initial point approximately coincides with the diagonal line of the rectangular core concrete in the middle zone. Taking the parabola along the width as an example (Fig. 3), the tangent line at the initial point is the diagonal line of rectangle ABCD, which is denoted as AC in the figure. Then, we have

$$\tan\alpha = \frac{h-2t}{b-2r_1-2t} \tag{12}$$

Similarly

$$\tan\beta = \frac{b-2t}{h-2r_1-2t} \tag{13}$$

Substituting Eqs. (12) and (13) into Eq. (11) yields

$$k_e = 1 - \frac{(h-2t)(b-2r_1-2t) + (b-2t)(h-2r_1-2t)}{3(b-2t)(h-2t) - 3(4-\pi)r_1^2} \tag{14}$$

Based on Fig. 1, when the corner radius equals zero, i.e., $r_0 = r_1 = 0$, the typical column turns into a rectangular column, and the effectiveness factor k_e in Eq. (14) is 1/3; when $r_0 = h/2 = b/2$, $r_1 = h/2 - t = b/2 - t$, the typical column turns into a circular column, and the effectiveness factor k_e in Eq. (14) is 1.0, which indicates that the concrete is effectively confined.

Unified Theoretical Model with Shape Effect Considered

The original CFST columns with different cross-sectional shapes were replaced by corresponding equivalent circular columns. Then, a unified theoretical model was developed, based on the established equivalent circular column, by considering the shape effect. Two parts, i.e., the models of steel tube and confined concrete, are included in the proposed unified theoretical model, which are introduced in the following subsections.

Modelling of Steel Tube

The stress state of the steel tube is indicated by von Mises stress σ_e. Considering that the steel tube in an axially loaded circular CFST column is subjected to a biaxial stress state, σ_e can be given by

$$\sigma_e = \sqrt{\sigma_{\text{sz},i-1}^2 - \sigma_{\text{sz},i-1}\sigma_{\text{sh},i-1} + \sigma_{\text{sh},i-1}^2} \tag{15}$$

where $\sigma_{\text{sz},i-1}$ and $\sigma_{\text{sh},i-1}$ = the axial and hoop stresses of the steel tube at the $i-1$-th loading step, respectively.

In the elastic stage ($\sigma_e < f_y$), the stress on the steel tube can be determined by the generalized Hooke law

$$\begin{Bmatrix} \mathrm{d}\sigma_{\text{sz},i} \\ \mathrm{d}\sigma_{\text{sh},i} \end{Bmatrix} = \frac{E_s}{1-\upsilon_s^2}\begin{bmatrix} 1 & \upsilon_s \\ \upsilon_s & 1 \end{bmatrix}\begin{Bmatrix} \mathrm{d}\varepsilon_{\text{sz},i} \\ \mathrm{d}\varepsilon_{\text{sh},i} \end{Bmatrix} \tag{16}$$

where $\mathrm{d}\varepsilon_{\text{z},i}$ and $\mathrm{d}\varepsilon_{\text{h},i}$ = the incremental axial and hoop strains, respectively; and E_s and ν_s = the modulus of elasticity and the Poisson's ratio of the steel tube, respectively. In this paper, E_s is taken as 200 GPa, and ν_s is taken as 0.283.

In the plastic stage ($\sigma_e \geq f_y$), the Prandtl-Reuss equation is adopted, given as follows:

$$\begin{Bmatrix} \mathrm{d}\sigma_{\text{sz},i} \\ \mathrm{d}\sigma_{\text{sh},i} \end{Bmatrix} = \frac{E_s}{1-\upsilon_s}\begin{bmatrix} 1-S_a^2/S_c & \upsilon_s - S_aS_b/S_c \\ \upsilon_s - S_aS_b/S_c & 1-S_b^2/S_c \end{bmatrix}\begin{Bmatrix} \mathrm{d}\varepsilon_{\text{sz},i} \\ \mathrm{d}\varepsilon_{\text{sh},i} \end{Bmatrix} \tag{17}$$

$$S_a = s_z + \upsilon_s s_h \tag{18}$$

$$S_b = s_h + \upsilon_s s_z \tag{19}$$

$$S_c = s_z^2 + s_h^2 + 2\upsilon_s s_z s_h \tag{20}$$

where s_z and s_h = the deviatoric stresses in the axial and hoop directions, respectively

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$$s_z = (2\sigma_{sz,i-1} - \sigma_{sh,i-1})/3 \tag{21}$$

$$s_h = (2\sigma_{sh,i-1} - \sigma_{sz,i-1})/3 \tag{22}$$

For a given lateral strain, another equation is required to determine the stresses on the steel tube. Assuming that the steel tube and the core concrete are perfectly bonded, we have $\varepsilon_{sz} = \varepsilon_{cz}$, and $\varepsilon_{sh} = \varepsilon_{ch}$. Additionally, based on Teng et al. (2007), the relationship between the axial and lateral strains of confined concrete can be indicated by

$$\frac{\varepsilon_{cz}}{\varepsilon_{c0}} = 0.85\left(8\frac{\sigma_r}{f_c} + 1\right)\left[\left(1 - 0.75\frac{\varepsilon_{ch}}{\varepsilon_{c0}}\right)^{0.7} - \exp\left(7\frac{\varepsilon_{ch}}{\varepsilon_{c0}}\right)\right] \tag{23}$$

where σ_r = lateral stress of confined concrete, and can be derived as

$$\sigma_r = -\frac{t_e}{R_e}\sigma_{sh} \tag{24}$$

Modelling of Confined Concrete

The stress-strain model of confined concrete proposed by Attard and Setunge (1996), which has been proved to have high accuracy with a wide range of concrete strengths, is modified to consider the shape effect. The modified model is given as

$$Y = \frac{AX + BX^2}{1 + (A-2)X + (B+1)X^2} \tag{25}$$

where $Y = \sigma_{cz}/f_{cc}$, $X = \varepsilon_{cz}/\varepsilon_{cc}$; σ_{cz} and ε_{cz} are the axial stress and strain of confined concrete, respectively; f_{cc} and ε_{cc} = peak stress of concrete and the corresponding strain under the lateral stress at a specified loading step, respectively; and A and B = coefficients which determine the shape of the stress-strain curve.

The peak stress of concrete was determined using the model of Richart et al. (1928), which is simple and found to yield good results for CFST columns (Lai and Ho 2016)

$$f_{cc} = f_c + 4.1\sigma_{re} \tag{26}$$

where $\sigma_{re}(= k_e\sigma_r)$ is the effective lateral stress; and k_e = confinement effectiveness factor introduced to consider the shape effect, which can be determined using Eq. (14).

The corresponding strain is calculated by the following equation, which was established based on experimental calibration using test results of CFST columns with different cross-sectional shapes:

$$\varepsilon_{cc} = \varepsilon_{c0}\left(1 + \frac{17.5}{k_e^{0.25}}\frac{\sigma_{re}}{f_c}\right) \tag{27}$$

When the shape factor $k_e = 1.0$ and 0.531, the proposed ε_{cc} equation turns into the models of Richart et al. (1928) and Teng et al. (2007), respectively; ε_{c0} = strain of unconfined concrete at the stress f_c, and can be obtained by the model of Wee et al. (1996)

$$\varepsilon_{c0} = 7.8 \times 10^{-4} f_c^{0.25} \tag{28}$$

The coefficients A and B in Eq. (25) determine the shape of the stress-strain curve of concrete. For the ascending branch, they are given by

$$A = \frac{E_c\varepsilon_{cc}}{f_{cc}} \tag{29}$$

$$B = \frac{(A-1)^2}{(1 - 0.45f_c/f_{cc})} \tag{30}$$

where E_c = modulus of elasticity of concrete, which is calculated using the recommended equation in ACI 318-19 (ACI 2019)

$$E_c = 4700\sqrt{f_c} \tag{31}$$

The coefficients A and B for the descending branch of the axial stress–strain curve are given as

$$A = \left(\frac{\varepsilon_2 - \varepsilon_1}{\varepsilon_{cc}}\right)\left[\frac{\varepsilon_2 f_1}{\varepsilon_1(f_{cc} - f_1)} - \frac{4\varepsilon_1 f_2}{\varepsilon_2(f_{cc} - f_2)}\right] \tag{32}$$

$$B = (\varepsilon_1 - \varepsilon_2)\left[\frac{f_1}{\varepsilon_1(f_{cc} - f_1)} - \frac{4f_2}{\varepsilon_2(f_{cc} - f_2)}\right] \tag{33}$$

$$\frac{f_1}{f_{cc}} = k_e^{0.18}\left[\frac{0.41 - 0.17\ln(f_c)}{5.06(\sigma_{re}/f_c) + 1} + 1\right] \tag{34}$$

$$\frac{f_2}{f_{cc}} = k_e^{0.16}\left[\frac{0.45 - 0.25\ln(f_c)}{6.35(\sigma_{re}/f_c)^{0.62} + 1} + 1\right] \tag{35}$$

$$\frac{\varepsilon_1}{\varepsilon_{cc}} = \frac{0.5 - 0.3\ln(f_c)}{1.12(\sigma_{re}/f_c)^{0.26} + 1} + 2 \tag{36}$$

$$\varepsilon_2 = 2\varepsilon_1 - \varepsilon_{cc} \tag{37}$$

where f_1 and ε_1 = stress and strain of concrete at the inflection point in the descending branch, respectively; and f_2 = stress at strain $\varepsilon_2 = 2\varepsilon_1 - \varepsilon_{cc}$. The confinement effectiveness factor k_e is incorporated into the models of A and B to consider the cross-sectional shape effect on the shape of load-deflection curves of concrete.

Implementation of Proposed Unified Theoretical Model

The implementation process of the proposed unified theoretical model is given in Fig. 4. Following the flowchart, the axial load-deflection curves of CFST columns with different shapes can be determined. The implementation steps are briefly given as follows:

Step 1: Input the column parameters and initialize the conditions, i.e., $\varepsilon_{sz,0} = \varepsilon_{sh,0} = 0$.

Step 2: Determine the radius of the core concrete and the thickness of the steel tube in the equivalent circular column, and the confinement effectiveness factor.

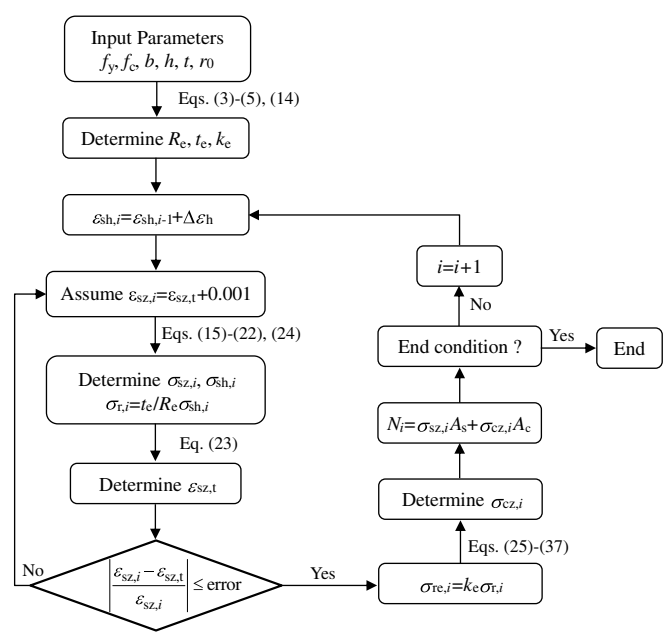

Fig. 4. Implementation flowchart of proposed unified theoretical method.

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Step 3: Give a hoop strain and assume an axial strain of the steel tube. Then, determine the axial and hoop stresses.

Step 4: Determine the axial stress of the core concrete.

Step 5: Obtain the axial load of the column by summing the parts of core concrete and steel tube.

Step 6: End the program if the end condition is satisfied; otherwise, increase the loading step and repeat Steps 3–5.

Verification of the Proposed Theoretical Model

The proposed unified theoretical model is verified in this section. Experimental test data of axially loaded CFST stub columns with different cross-sectional shapes, assembled from the literature, are used for the model verification. Details of the collected test databases are given in the following subsection.

Test Database

In the past decades, numerous experimental tests have been conducted to investigate the behavior of CFST columns with different cross-sectional shapes subjected to axial compression. A total of 982 specimens, including 550 circular columns, 396 rectangular (and square) columns with and without corner radius, and 36 round-ended columns were collected from the literature by the authors of the present paper, which was used for model verification. The details of assembled circular CFST columns (not given in the present paper) can be seen in Lin et al. (2020b), whilst those of collected rectangular (and square) and round-ended CFST columns are summarized in Tables 2 and 3, respectively.

The collected test databases cover a wide range of column parameters. The circular CFST columns were collected from 47 published papers, and have 176 MPa $\leq f_y \leq$ 835 MPa, 15 MPa $\leq f_c \leq$ 185.6 MPa, 60 mm $\leq b \leq$ 1,020 mm, 60 mm $\leq h \leq$ 1,020 mm, and $13.4 \leq b/t \leq 319.2$; whilst the rectangular (and square) CFST columns were collected from 27 published papers, and have 194 MPa $\leq f_y \leq$ 835 MPa, 11.2 MPa $\leq f_c \leq$ 156.8 MPa, 60 mm $\leq b \leq$ 750 mm, 60 mm $\leq h \leq$ 750 mm, and $1.0 \leq b/h \leq 2.0$, $5.4 \leq b/t \leq 135.8$. Compared with circular or rectangular CFST columns, research on round-ended CFST columns is relatively limited. A total of 36 specimens from four papers were collected. The ranges of the parameters are: 299.5 MPa $\leq f_y \leq$ 382 MPa, 31.43 MPa $\leq f_c \leq$ 89.4 MPa, 120 mm $\leq b \leq$ 806 mm, 50 mm $\leq h \leq$ 264 mm, $1.17 \leq b/h \leq 4.24$, and $51.2 \leq b/t \leq 219.6$.

The reported concrete strengths in previous studies were based on different test standards, which were all converted to the standard concrete strength f_c (150 × 300 mm cylinder) using the conversion method given in Lin et al. (2021). The collected test databases of CFST columns with different cross-sectional shapes were used to verify the performance of the proposed model through the comparison of axial compressive strengths and load-deflection curves between experimental and predicted results.

Axial Load-Carrying Capacity

The axial load-carrying capacity predicted by the proposed model are compared with experimental results in Fig. 5. The mean value

Table 2. Collected rectangular (and square) CFST columns with and without corner radius

Reference	Specimen numbers	f_y (MPa)	f_c (MPa)	b (mm)	h (mm)	t (mm)	r_0 (mm)	b/h
Uy (1998)	5	300	32–50	126–306	126–306	3	0	1
Yamamoto et al. (2000)	8	300–395	24.6–60.9	100.1–300.7	100.1–300.7	2.18–6.1	0	1
Uy (2001a)	6	784.2	28–32	110–210	110–210	5	0	1
Uy (2001b)	2	784.2	50	110	110	5	0	1
Han et al. (2001)	20	321–330	11.2–44.1	120–200	120–200	3.84–5.86	0	1
Han (2002)	24	194–228	50.8	90–160	70–135	2.86–7.6	0	1–1.75
Han and Yao (2003)	8	340.1	17.9	135–360	90–240	2.65	0	1.5–2
Liu et al. (2003)	21	550	61.7–72.6	100.3–200.2	80.1–181.2	4.18	0	1–2
Han and Yao (2004)	6	303.5	50	200	200	3	0	1
Sakino et al. (2004)	48	262–835	24.3–87.1	119–324	119–324	4.38–9.45	0	1
Liu and Gho (2005)	26	300–495	55–106	120–220	90–200	4–5.8	0	1–2
Liu (2005)	22	495	60–89	106–190	80–140	4	0	1–2
Han et al. (2005)	24	282–404	42.9–71.5	60–250	60–250	1.87–2	0	1
Zhang et al. (2005)	50	255.1–347.3	48.7–66.8	101.3–194.8	100.7–143.1	2.81–5.15	0	1–1.6
Lam and Gardner (2008)	6	385–458	30–74	99.8–101.3	99.3–101.6	2–5	0	1
Lee et al. (2008)	2	539	40	160–240	160–240	8	0	1
Chen et al. (2012)	4	358–389	42.5	410–500	410–500	10–16	0	1
Uy et al. (2013)	8	760	80–98	100–200	100–200	5	0	1
Aslani et al. (2015)	12	701	21–54.5	70–200	70–200	5	0	1
Du et al. (2016b)	6	488.38	35.8–47	150–203	101–150	8.275	0	1–2
Du et al. (2016a)	8	423.2–514.5	35.8–47	120–182	100–122	5.7–7.69	0	1.17–1.5
Zhu et al. (2017)	6	381.7–437.9	19.1–20.9	197–201	197–201	6.1–10.3	0	1
Xiong et al. (2017)	15	446–779	140.5–156.8	150	150	8–12.5	0	1
Chen et al. (2018)	9	306.7–371.6	61.8–125	100.3–100.7	100.3–100.7	2.03–7.63	0	1
Ibañez et al. (2018)	6	270.8–342.6	40.4–94.3	125–150	100–125	3–4	0	1–1.5
Wu et al. (2018)	6	302–334	27.9	300–750	300–750	6–15	0	1
Yan et al. (2019)	16	444.6–668.8	88–125	100	100	4.9–18.5	0	1
Tomii and Sakino (1979)	8	194.2–339.3	18.9–30.6	100	100	2.2–4.25	4.6–9.3[a]	1
Young and Ellobody (2006)	14	448–536	46.6–83.5	140.2–197.6	80–150.6	2.78–5.84	4.78–9.83[a]	1–2
Minimum	—	194	11.2	60	60	1.87	0	1
Maximum	—	835	156.8	750	750	18.5	9.83	2

[a]Inner radii of steel tube were provided in original papers. In the table, outer radii are given, which are obtained using Eq. (5).

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Table 3. Collected round-ended CFST columns

Reference	Specimen numbers	f_y (MPa)	f_c (MPa)	b (mm)	h (mm)	t (mm)	r_0 (mm)	b/h
Gu et al. (2014)	10	299.5–327.7	33.2–42.4	299.3–396	249–264	3.72–5.92	124.5–132	1.17–1.52
Ding et al. (2015)	22	299.5–327.7	32.2–49	299–806	186–264	3.67–5.92	93–132	1.17–4.24
Ding et al. (2017)	2	382	31.43	500–505	251–252	3.87–3.91	125.5–126	1.98–2.01
Piquer et al. (2019)	2	306.98	40.8–89.4	120	50	2	25	2.4
Minimum	—	299.5	31.43	120	50	2	25	1.17
Maximum	—	382	89.4	806	264	5.92	132	4.24

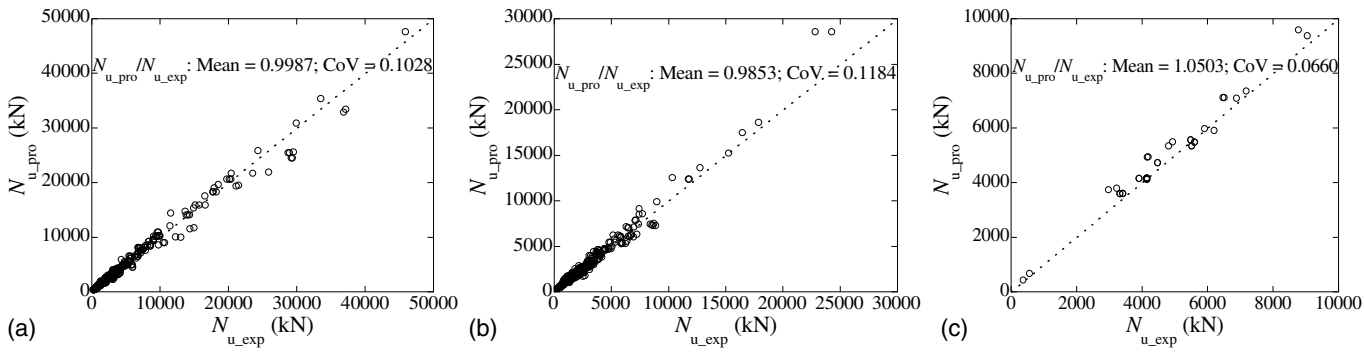

Fig. 5. Comparison of axial load-carrying capacity between experimental results and predicted ones for CFST columns with different shapes: (a) circular; (b) rectangular and square; and (c) round-ended.

and coefficient of variation (CoV) of $N_{u_\mathrm{pro}}/N_{u_\mathrm{exp}}$ for circular CFST columns are 0.9987 and 0.1028, respectively, whilst those for rectangular (and square) CFST columns are 0.9853 and 0.1184, respectively, where N_{u_pro} is the capacity predicted by the proposed model and N_{u_exp} is that obtained from experimental tests. Results suggest that the axial load-carrying capacity of circular and rectangular (and square) CFST columns are well predicted by the proposed unified theoretical model. The mean value and CoV of $N_{u_\mathrm{pro}}/N_{u_\mathrm{exp}}$ for round-ended CFST columns are 1.0503 and 0.0660, respectively. It seems that the proposed model slightly overestimates the axial load-carrying capacity of the round-ended CFST columns collected in the present study. However, it is worth noting that the amount of experimental test data for round-ended CFST columns is relatively limited. Because experimental results generally exhibit great data scatter, the accuracy of the proposed model seems acceptable and may be improved if more test data are available in the future.

The axial load-carrying capacity for all collected CFST columns is given in Fig. 6. The mean and CoV of $N_{u_\mathrm{pro}}/N_{u_\mathrm{exp}}$ are 0.9952 and 0.1088, respectively. This suggests that the predicted results of the proposed model agree quite well with the experimental results.

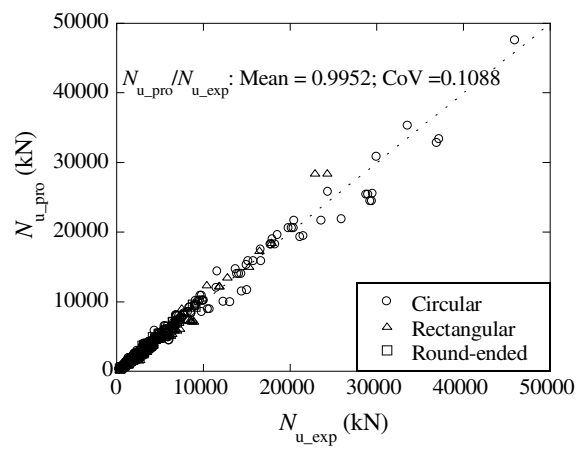

Fig. 6. Comparison of axial load-carrying capacity between experimental results and predicted ones for all collected CFST columns.

Load-Deflection Curves

The performance of the proposed unified theoretical model is further verified in this section through the comparison of load-deflection curves between experimental and predicted results.

The predicted load-deflection curves of circular CFST columns are compared with experimental results in Fig. 7. Four specimens from different studies, i.e., specimens 490-36-31WL (Lin et al. 2018), CA2 (Han et al. 2005), CC8-A-8 (Sakino et al. 2004), and C15 (Xiong et al. 2017), were used to assess the performance of the proposed theoretical model. The theoretical model of Teng et al. (2013) was used for model comparison. Figs. 7(a and b) show the results of circular CFST columns using normal concrete and self-compacting concrete, respectively. The results suggest that both the proposed model and Teng et al.'s model predict the results with reasonable accuracy. The specimen in Fig. 7(b) has concrete strength of 75.6 MPa, and its post-peak load-deflection curve is well-simulated by the proposed model, suggesting that the proposed model is able to predict the decay of the post-peak curve for specimens with high strength concrete. Figs. 7(c and d) show the results of circular CFST columns using ultra-high strength steel ($f_y = 853$ MPa) and ultra-high strength concrete ($f_c = 155.8$ MPa), respectively. For the specimen using ultra-high strength steel, the predicted results of both Teng et al.'s model and the proposed model agree well with the experimental result. For the specimen using ultra-high strength concrete, the proposed model predicts the ascending branch quite well and yields a more accurate descending branch than Teng et al.'s model. Although the sudden drop behavior at the descending branch is not well captured by the proposed model, it is simulated with reasonable accuracy.

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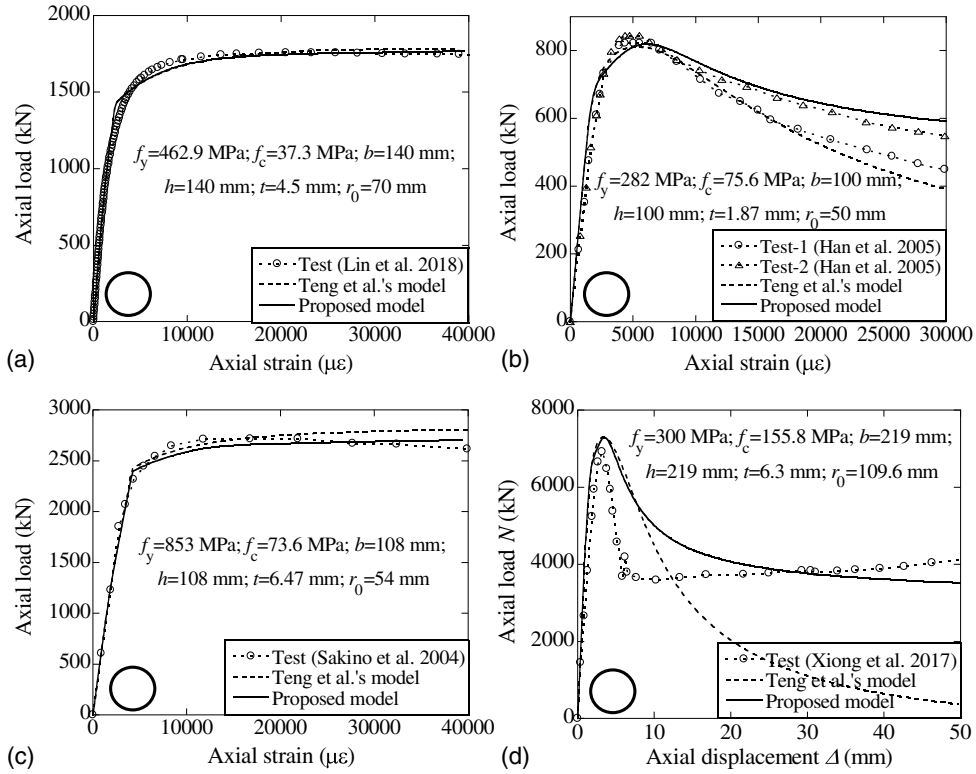

Fig. 7. Accuracy of proposed theoretical model for predicting load-deflection curves of circular CFST columns: (a) specimen 490-36-31WL; (b) specimen CA2; (c) specimen CC8-A-8; and (d) specimen C15.

The predicted load-deflection curves of rectangular (and square) CFST columns are shown in Fig. 8. Four columns, i.e., specimen sczs1-1-4 (Han et al. 2001), A300 (Wu et al. 2018), and rc7 and rc12 (Han 2002), were used for the model assessment. The existing models for rectangular (and square) columns were essentially empirically established. Although empirical models are beyond the scope of this paper, to further verify the performance of the proposed unified theoretical model, the empirical model of Tao et al. (2021) was adopted for model comparison. Figs. 8(a and b) show the results of the specimens with aspect ratio $b/h = 1.0$ (square columns), while Figs. 8(c and d) show the results of the specimens with aspect ratios $b/h = 1.33$ and 1.75, respectively (rectangular columns). Results suggest that Tao et al.'s model well predicts the ascending branch but slightly underestimate the descending branch, while the proposed theoretical model generally makes good predictions. The proposed theoretical model can capture stress-strain responses of rectangular columns with wide concrete strengths mainly because: (1) The proposed unified theoretical model was developed based on a well-established equivalent circular CFST column; and (2) the concrete model of Attard and Setunge (1996), which has been verified to be applicable to a broad range of concrete strengths, is adopted in the proposed theoretical model.

The performance of the proposed theoretical model for round-ended CFST columns is assessed in Fig. 9 by using the specimens WST2 and WST5 that were tested by Ding et al. (2015). Because no theoretical models exist for round-ended CFST columns, the empirical model of Patel (2020) was used for model comparison. From the figure, it can be seen that Patel's model predicts the ascending branch of the load-deflection curve well but underestimates the peak load and the descending branch, while the proposed theoretical model generally yields good results for both the ascending and descending branches.

The foregoing discussion suggests that the proposed unified theoretical model is capable of accurately predicting the load-carrying capacity and load-deflection curves of axially-loaded CFST columns with different cross-sectional shapes. It should be noted that the calculated results slightly differ from the test ones, because: (1) Any model has errors, and (2) experimental results generally exhibit large scatter.

Unified Design Model for CFST Columns with Different Shapes

In this section, a unified design model of axial load-carrying capacity is developed based on the proposed theoretical model. The axial load-carrying capacity of CFST columns is related to the mechanical properties of its materials, i.e., f_y and f_c, and the geometrical properties of columns, i.e., t, r_0, h, and b. Among the geometrical properties, parameters r_0, h, and b determine the shape of the column. That is to say, if the radius ratio (r_0/h) and the aspect ratio (b/h) are known, the cross-sectional shape of the column can be determined. Several typical cases are given as follows:

- Circular: $r_0/h = 1/2$ and $b/h = 1.0$.
- Round-ended: $r_0/h = 1/2$ and $b/h > 1.0$.
- Square: $r_0/h = 0$ and $b/h = 1.0$.
- Rectangular: $r_0/h = 0$ and $b/h > 1.0$.
- Rectangular column with corner radius: $0 < r_0/h < 1/2$ and $b/h > 1.0$.

The foregoing discussion suggests that for CFST columns with different cross-sectional shapes that have same f_y, f_c, and t, their differences in axial load-carrying capacity are attributed to different r_0/h and b/h. Assuming the influences of r_0/h and b/h on the axial load-carrying capacity of CFST columns are independent, and taking the sectional capacity of circular CFST columns as the

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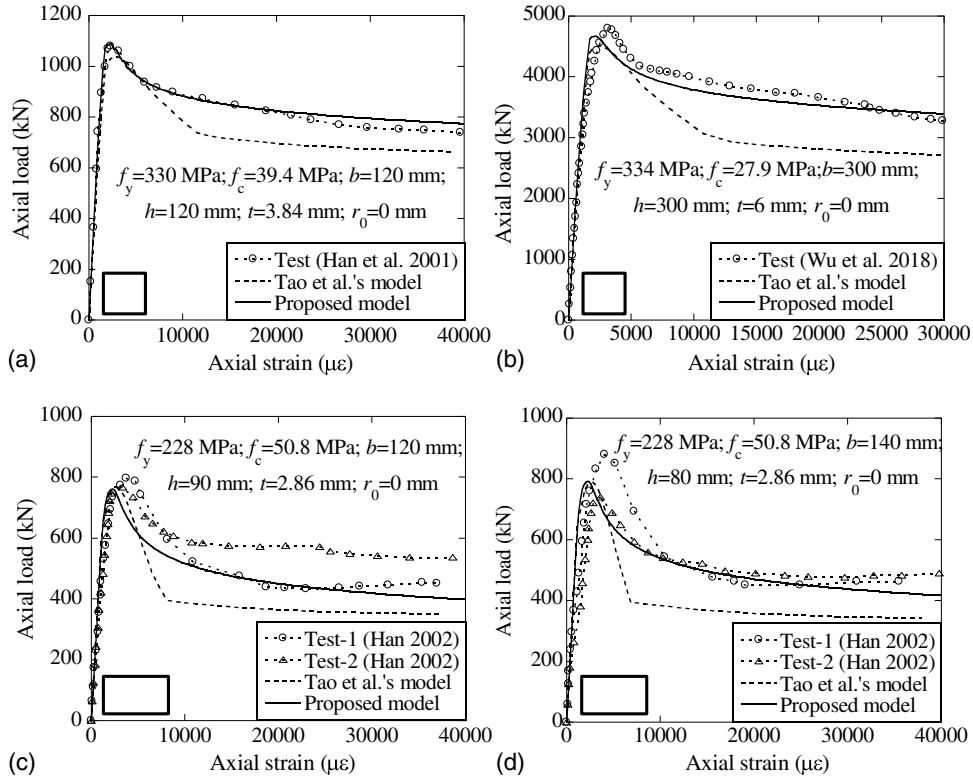

Fig. 8. Accuracy of proposed theoretical model for predicting load-deflection curves of rectangular CFST columns: (a) specimen sczs1-1-4; (b) specimen A300; (c) specimen rc7; and (d) specimen rc12.

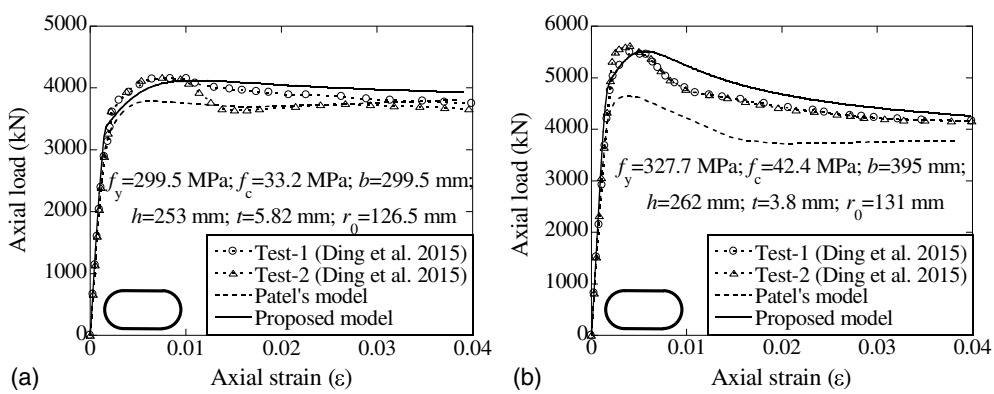

Fig. 9. Accuracy of proposed theoretical model for predicting load-deflection curves of round-ended CFST columns: (a) specimen WST2; and (b) specimen WST5.

reference, then the axial load-carrying capacity of CFST columns with different cross-sectional shapes can be expressed as follows:

$$N_u = \varphi_r\left(\frac{r_0}{h}\right)\varphi_a\left(\frac{b}{h}\right)N_{\mathrm{uc}} \tag{38}$$

where N_{uc} = section capacity of the circular CFST column (reference column), whose diameter (D) equals the cross-sectional height (h); and ϕ_r and ϕ_a = coefficients that indicate the influences of the radius ratio (r_0/h) and aspect ratio (b/h) on the different axial strengths between circular and other shapes, respectively. Note that r_0/h and b/h are within the intervals of $0 \le r_0/h \le 0.5$ and $b/h \ge 1.0$, respectively. Additionally, because Eq. (38) is also applicable to circular CFST columns, the coefficients φ_r and φ_a should have $\varphi_r(0.5) = \varphi_a(1) = 1.0$.

The sectional capacity of circular CFST columns N_{uc} in Eq. (38) can be predicted by the model proposed by the authors (Lin et al. 2020b), which is given as

$$N_{\mathrm{uc}} = f_{\mathrm{cm}}A_{\mathrm{cc}} + f_{\mathrm{sm}}A_{\mathrm{sc}} \tag{39}$$

where $A_{\mathrm{cc}} = [\pi(D/2 - t)^2]$ and $A_{\mathrm{sc}} = [\pi(D/2)^2 - A_{\mathrm{cc}}]$ are the sectional areas of concrete and steel tube of circular CFST columns, respectively; and f_{cm} and f_{sm} = stresses corresponding to the maximum load of CFST columns for concrete and steel tube, respectively, expressed as

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$$\frac{f_{\mathrm{cm}}}{f_c} = R_{\mathrm{d,cc}}\left[1 + 52 f_y^{0.1} f_c^{-0.85}\left(\frac{D}{t}\right)^{-0.6}\right] \tag{40}$$

$$\frac{f_{\mathrm{sm}}}{f_y} = 1.53 f_y^{-0.05}\left(1 - 0.002\frac{D}{t}\right) \tag{41}$$

where $R_{\mathrm{d,cc}}$ = reduction factor to consider the size effect of confined concrete, which is given as

$$R_{\mathrm{d,cc}} = 1.67^{(1+\eta)^{-8}}(D-2t)^{-(0.112\times 0.0005^{\eta})} \tag{42}$$

where η = confinement coefficient, defined as

$$\eta = \frac{2t}{D-2t}\frac{f_y}{f_c} \tag{43}$$

The derivation detail of size effect involved in the design formula of circular CFST columns can be seen in Lin and Teng (2020), and is not repeated in the present paper for simplicity.

To determine the coefficient φ_r, three groups of specimens, including 35 columns, were designed, details of which are given in Table 4. The load-carrying capacities of all designed columns (N_u) were calculated using the proposed theoretical model, which are also given in Table 4. N_u/N_{uc} is the normalized load-carrying capacity with respect to that of a corresponding circular CFST column (reference column). Based on Eq. (38), N_u/N_{uc} can be given as

$$\frac{N_u}{N_{\mathrm{uc}}} = \varphi_r\left(\frac{r_0}{h}\right)\varphi_a\left(\frac{b}{h}\right) \tag{44}$$

All columns in Table 4 are designed to have a b/h of 1.0, and hence coefficient φ_a equals 1.0. Then, based on Eq. (44), N_u/N_{uc} is φ_r. The influence of column parameters on the relationships between φ_r and r_0/h is given in Fig. 10. It is interesting to note that:

- The φ_r versus r_0/h relation is significantly affected by the concrete strength but barely affected by the steel yield strength and the h/t ratio.
- For columns with normal strength concrete ($f_c = 30$ MPa), φ_r approximately equals unity despite the variations in the r_0/h ratio, whereas for the columns with high and ultrahigh strength concrete, φ_r decreases with and increasing r_0/h ratio.

Increasing the r_0/h ratio decreases the cross-sectional area and hence results in lower load-carrying capacity of CFST columns, but increasing the r_0/h ratio improves the confinement effect of concrete and hence leads to a higher load-carrying capacity. The results in Fig. 10 suggest that, when increasing the r_0/h ratio of CFST columns, the increase of load-carrying capacity due to higher confinement effects almost compensates for the decrease in load-carrying capacity due to the loss of cross-sectional area, except for the CFST columns with high or ultrahigh strength concrete. Moreover, the finding of the second point in the prior paragraph and the result in Fig. 10(a), suggest that, compared with the CFST column

Table 4. Designed specimens for determination of coefficient φ_r

Group	Specimens	f_y (MPa)	f_c (MPa)	b (mm)	h (mm)	t (mm)	r_0 (mm)	r_0/h	b/h	N_u (kN)	N_u/N_{uc}
One	S2-c1-t2-1[a]	450	30	150	150	3	75	0.5	1	1,408.3	1.00
	S2-c1-t2-2	450	30	150	150	3	56.25	0.375	1	1,420.4	1.01
	S2-c1-t2-3	450	30	150	150	3	37.5	0.25	1	1,424.6	1.01
	S2-c1-t2-4	450	30	150	150	3	18.75	0.125	1	1,423.4	1.01
	S2-c1-t2-5	450	30	150	150	3	0	0	1	1,418.4	1.01
	S2-c2-t2-1[a]	450	90	150	150	3	75	0.5	1	2,433.8	1.00
	S2-c2-t2-2	450	90	150	150	3	56.25	0.375	1	2,539.9	1.04
	S2-c2-t2-3	450	90	150	150	3	37.5	0.25	1	2,607.5	1.07
	S2-c2-t2-4	450	90	150	150	3	18.75	0.125	1	2,658.6	1.09
	S2-c2-t2-5	450	90	150	150	3	0	0	1	2,666.7	1.10
	S2-c3-t2-1[a]	450	150	150	150	3	75	0.5	1	3,401.9	1.00
	S2-c3-t2-2	450	150	150	150	3	56.25	0.375	1	3,608.2	1.06
	S2-c3-t2-3	450	150	150	150	3	37.5	0.25	1	3,772.9	1.11
	S2-c3-t2-4	450	150	150	150	3	18.75	0.125	1	3,886.3	1.14
	S2-c3-t2-5	450	150	150	150	3	0	0	1	3,910.8	1.15
Two	S1-c1-t2-1[a]	250	30	150	150	3	75	0.5	1	1,051.6	1.00
	S1-c1-t2-2	250	30	150	150	3	56.25	0.375	1	1,064.9	1.01
	S1-c1-t2-3	250	30	150	150	3	37.5	0.25	1	1,069	1.02
	S1-c1-t2-4	250	30	150	150	3	18.75	0.125	1	1,066.2	1.01
	S1-c1-t2-5	250	30	150	150	3	0	0	1	1,066.4	1.01
	S3-c1-t2-1[a]	650	30	150	150	3	75	0.5	1	1,732.4	1.00
	S3-c1-t2-2	650	30	150	150	3	56.25	0.375	1	1,770.7	1.02
	S3-c1-t2-3	650	30	150	150	3	37.5	0.25	1	1,782.2	1.03
	S3-c1-t2-4	650	30	150	150	3	18.75	0.125	1	1,787	1.03
	S3-c1-t2-5	650	30	150	150	3	0	0	1	1,745.5	1.01
Three	S2-c1-t1-1[a]	450	30	150	150	1.5	75	0.5	1	1,022.3	1.00
	S2-c1-t1-2	450	30	150	150	1.5	56.25	0.375	1	1,039.5	1.02
	S2-c1-t1-3	450	30	150	150	1.5	37.5	0.25	1	1,050.5	1.03
	S2-c1-t1-4	450	30	150	150	1.5	18.75	0.125	1	1,056	1.03
	S2-c1-t1-5	450	30	150	150	1.5	0	0	1	1,053.9	1.03
	S2-c1-3-1[a]	450	30	150	150	4.5	75	0.5	1	1,736.7	1.00
	S2-c1-t3-2	450	30	150	150	4.5	56.25	0.375	1	1,765.6	1.02
	S2-c1-t3-3	450	30	150	150	4.5	37.5	0.25	1	1,774.9	1.02
	S2-c1-t3-4	450	30	150	150	4.5	18.75	0.125	1	1,777.6	1.02
	S2-c1-t3-5	450	30	150	150	4.5	0	0	1	1,773.4	1.02

[a]Reference column.

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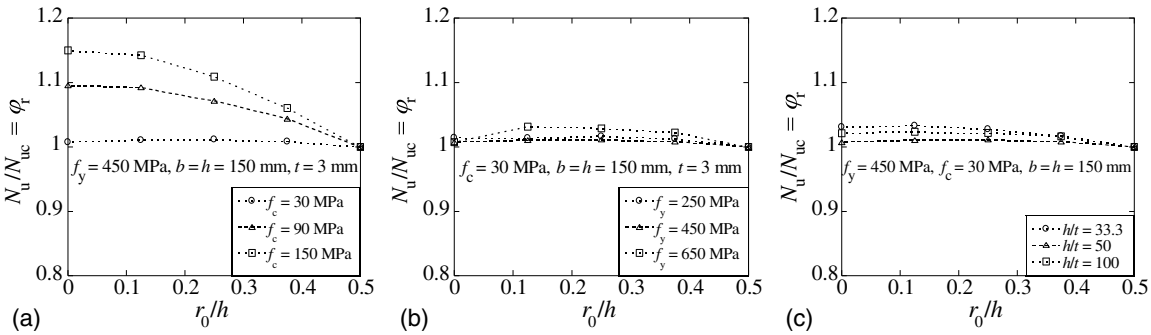

Fig. 10. Influence of column parameters on relationship between φ_r and r_0/h ($b/h = 1.0$, $\varphi_a = 1.0$): (a) concrete compressive strength; (b) steel yield strength; and (c) h/t ratio.

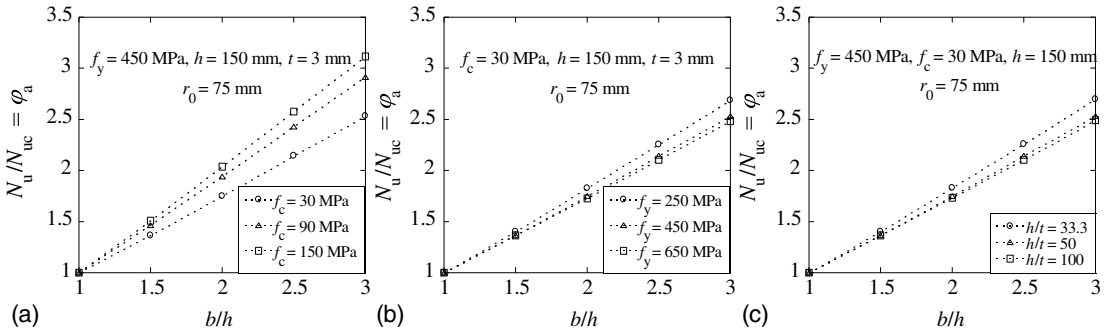

Fig. 11. Influence of column parameters on relationship between φ_a and b/h ($r_0/h = 0.5$, $\varphi_r = 1.0$): (a) concrete compressive strength; (b) steel yield strength; and (c) h/t ratio.

with normal strength concrete, those with higher strength concrete benefit less from confinement effects. This is because higher strength concrete exhibits lower lateral expansion. Based on the results of the parametric study, a simple model for coefficient φ_r is proposed as follows:

$$\varphi_r = 1 + 0.003 f_c[1 - \exp(r_0/h - 0.5)] \tag{45}$$

Similarly, the influence of column parameters on the φ_a versus b/h relationship were investigated and are shown in Fig. 11. The specimens designed for the determination of φ_a have a r_0/h ratio of 0.5, hence $\varphi_r = 1.0$ and $N_u/N_{\mathrm{uc}} = \varphi_a$. From Fig. 11, it can be observed that φ_a increases almost linearly with increasing b/h. Moreover, compared to concrete strength, the influence of steel yield strength and h/t ratio is marginal, which hence is ignored in this paper. A simple model of coefficient φ_a is then proposed as

$$\varphi_a = 1 + (0.002 f_c + 0.62)(b/h - 1) \tag{46}$$

The proposed models for coefficients φ_r and φ_a are compared with the results obtained by the proposed theoretical model in Fig. 12, which suggests that the coefficients are well predicted. A unified design model of load-carrying capacity for CFST columns with different cross-sectional shapes is then obtained by substituting Eqs. (39), (45), and (46) into Eq. (38). The collected experimental results were again adopted to assess the accuracy of the proposed design model for CFST columns with different cross-sectional shapes, which is shown in Fig. 13. The mean value and CoV of $N_{\mathrm{u_pro}}/N_{\mathrm{u_exp}}$ are 1.0001 and 0.1042, respectively, which suggests that the proposed design can accurately predict the load-carrying capacity of CFST columns with different cross-sectional shapes.

Conclusions

This paper proposes a unified theoretical model for axially loaded CFST columns with different cross-sectional shapes. The following conclusions can be drawn:

- An equivalent circular column was established for the axially loaded CFST column with different cross-sectional shapes (original columns) by assuming that they have the same cross-sectional areas as the core concrete and steel tube. By doing so, the differences in behaviors between the equivalent circular CFST column and the original column with another shape can be attributed to the shape effect.
- The confinement effectiveness of core concrete in CFST columns with different cross-sectional shapes was investigated, and a model of confinement effectiveness factor for considering the shape effect was proposed. A unified theoretical model was then developed for axially loaded CFST columns with different cross-sectional shapes, based on the established equivalent circular specimen with the shape effect considered.
- A test database of 982 CFST columns with different cross-sectional shapes, including 550 circular columns, 396 rectangular (and square) columns, and 36 round-ended columns, was collected for the model assessment. Good accuracy of the proposed theoretical model was verified by comparison with the experimental results.
- The proposed theoretical model was used to conduct parametric studies on the axial load-carrying capacity of CFST columns with different cross-sectional shapes. When increasing the r_0/h ratio of CFST columns with normal strength concrete, the increase in load-carrying capacity due to the higher confinement effect approximately compensates for the decrease of load-carrying capacity due to the loss of cross-sectional area.

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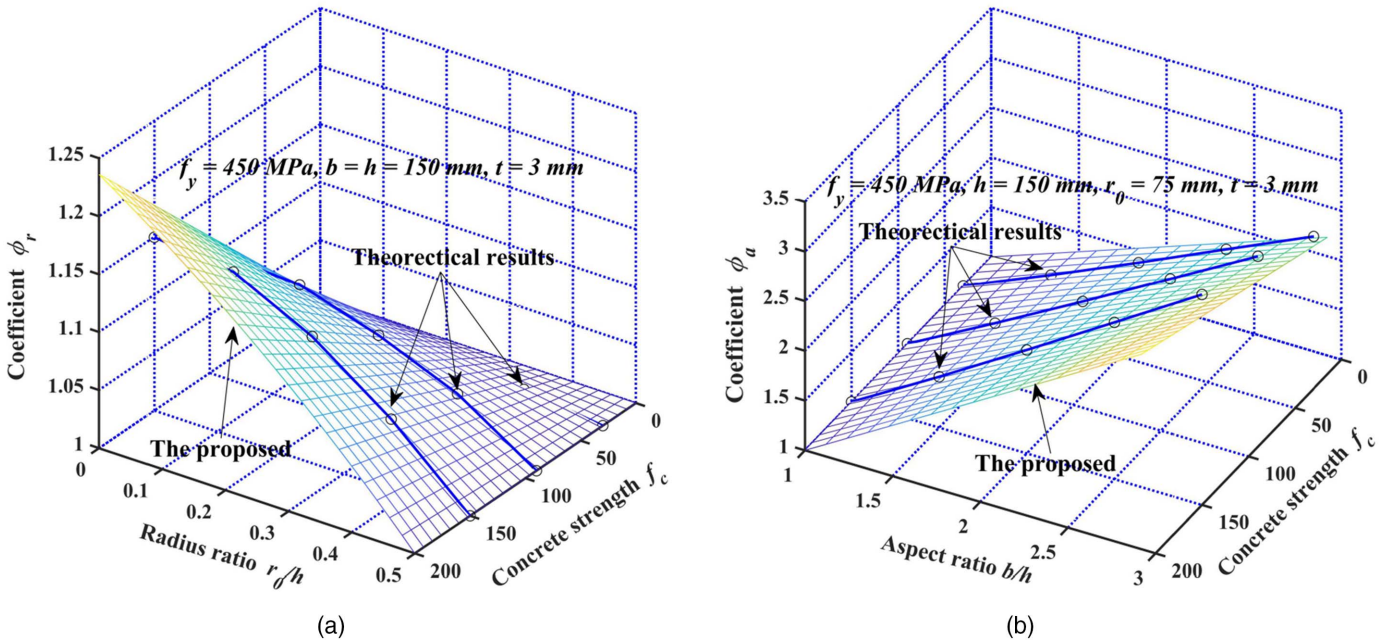

Fig. 12. Comparison between proposed models and theoretical results: (a) coefficient φ_r; and (b) coefficient φ_a.

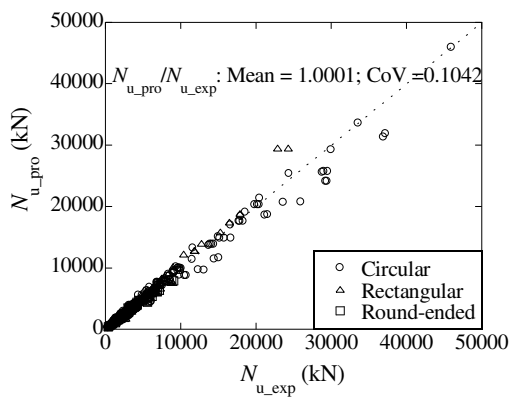

Fig. 13. Accuracy of proposed design model for predicting load-carrying capacity of CFST columns with different cross-sectional shapes.

- A unified design model applicable to CFST columns with different cross-sectional shapes was developed based on the results of parametric studies using the proposed theoretical model.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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